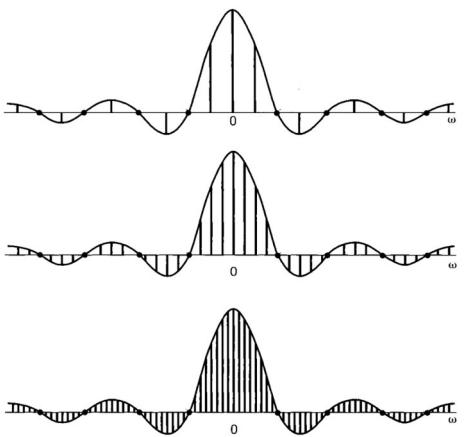
Fourier Transform

Back to Rectangular Wave

- Rectangular wave with D x 100% duty cycle, $z(t) = \sum \frac{\sin k\pi D}{k\pi} e^{jk\omega t}$
  - As T increases (equivalent to decrease in D), w decreases
    - Adds more frequency components (harmonics)



Fourier Analysis of Aperiodic Signals

- Changing granularity of frequency of the phasors:
  - $w \rightarrow \varepsilon$ (infinitesimal)
  - Spectrum becomes dense:
    - Energy spectral density: $|Z(jw)|^2$

$$z(t) = \sum \frac{1}{T}\left(\int_T z(t)e^{-jk\omega t}dt\right)e^{jk\omega t}$$
$$= \sum \frac{\omega}{2\pi} Z(jk\omega)e^{jk\omega t}$$
$$\lim_{T\to\infty} z(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} Z(j\nu)e^{j\nu t}d\nu$$

- Extending the inner product for infinite period-length:

$$Z(jk\omega) = \int_T z(t)e^{-jk\omega t}dt$$
$$\lim_{T\to\infty} Z(j\omega) = \int_{-\infty}^{\infty} z(t)e^{-j\omega t}dt$$

Projection for Aperiodic Signals

- Tackling infinite energy with Dirac delta function:

$$z(t) = \int a_\nu e^{j\nu t} d\nu$$

$$\lim_{T\to\infty} \int_T z(t) e^{-j\omega t} dt = \int a_\nu 2\pi\delta(\nu - \omega) d\nu$$

$$Z(j\omega) = a_\omega 2\pi$$

- Parseval's theorem: $\int |z(t)|^2 dt = \int \int a_\nu e^{j\nu t} d\nu \int a_\omega^* e^{-j\omega t} d\omega \;\; dt = \frac{1}{2\pi} \int |Z(j\nu)|^2 d\nu$

Fourier Transform

- Fourier transform of $x(t)$ is $X(jw)$ and the inverse Fourier transform of $X(jw)$ is $x(t)$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$$

- Parseval's theorem: $E_x = \int |x(t)|^2 dt = \frac{1}{2\pi}\int |X(j\omega)|^2 d\omega = \frac{E_X}{2\pi}$

The Sinc Function

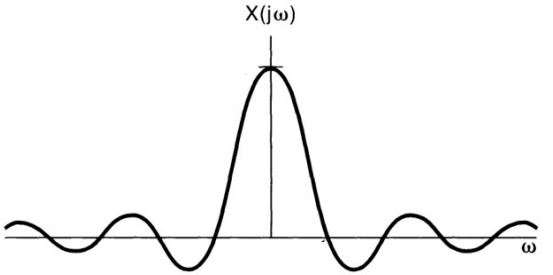
- The FT of rect(t/T_0) is $2(\sin wT_0/2)/w = T_0 \,\mathrm{sinc}\,(wT_0/2\pi)$

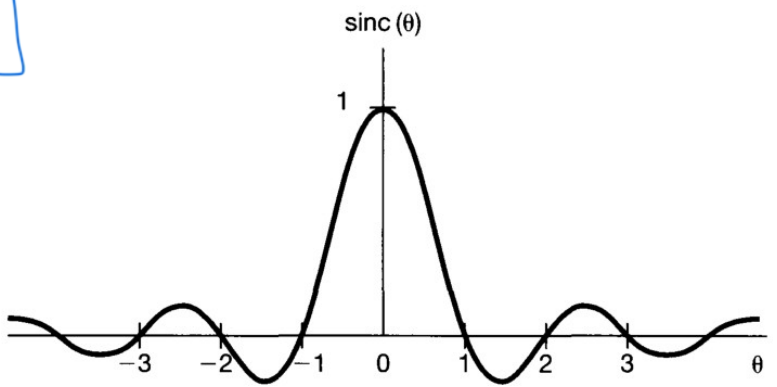
$1, t \in [-T_0/2, T_0/2]$

$$X(j\omega) = \int_{-T_0/2}^{T_0/2} e^{-j\omega t}\,dt = \frac{1}{j\omega_0}\left[e^{-j\omega T_0/2} + e^{j\omega T_0/2}\right]$$

$$= \frac{2}{\omega_0}\sin\left(\frac{\omega_0 T_0}{2}\right)$$

$$= T_0 \,\mathrm{sinc}\,(\omega T_0/2\pi)$$





The Uncertainty Principle

$x(t) \rightarrow X(\omega)$

$X(t) \rightarrow 2\pi x(-\omega)$

$x(-t) \rightarrow X(\omega)$

$\text{rect}(t/T_0) \rightarrow 2/\omega \sin\frac{\omega T_0}{2}$

- Find the FT of rect(t/T_0) and the IFT of rect(w/W_0)

$$X(\omega) = 1 \quad \omega \in [-\omega_0/2, \omega_0/2]$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi}\int_{-\omega_0/2}^{\omega_0/2} e^{j\omega t} d\omega = \frac{1}{2\pi jt}\left(e^{j\omega t/2} - e^{-j\omega t/2}\right)$$

$$x(t) = \frac{\sin(\omega_0 t/2)}{\pi t}$$

- Finite time length → infinite bandwidth & finite bandwidth → infinite time length

$$\text{IFT}\ \ \text{rect}(\omega/T_0) = -\frac{2}{2\pi}\frac{1}{t}\sin\left(\frac{-tT_0}{2}\right)$$

Fourier Transform - Properties

$x(t)$ – Time domain	$X(jw)$ – Frequency domain	Bandwidth
$x^*(t)$	$X^*(-jw)$	No change
$x(-t)$	$X(-jw)$	No change
$x(t-s)$	$X(jw)e^{-jws}$	No change (Only phase change in the other domain)
$x(t)e^{jvt}$	$X(j(w-v))$	
$x(ct)$	$X(jw/c) / \lvert c \rvert$ (density changes, area is fixed)	c > 1: expands c < 1: shrinks

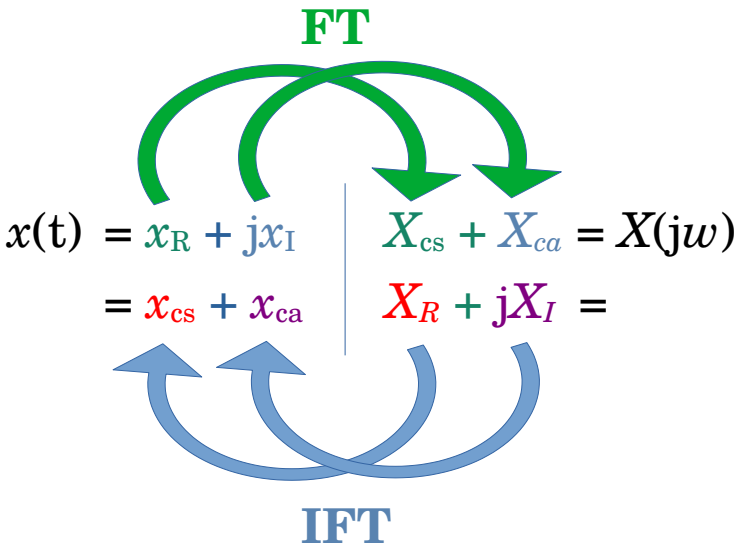
$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$X^*(j\omega) = \int_{-\infty}^{\infty} x^*(t) e^{j\omega t} dt \Rightarrow X^*(-j\omega) = \int_{-\infty}^{\infty} x^*(t) e^{-j\omega t} dt$$

$$\int_{-\infty}^{\infty} x(t-s) e^{-j\omega t} dt = \int_{-\infty}^{\infty} x(t-s) e^{-j\omega(t-s)} dt \cdot e^{-j\omega s} = X(j\omega) e^{-j\omega s}$$

Fourier Transform - Properties

$x(t)$ – Time domain	$X(jw)$ – Frequency domain
Even	Even
Odd	Odd
Conj. Sym.	Real
Conj. Asym.	Imaginary
Real	Conj. Sym.
Imaginary	Conj. Asym.



Duality of FT

- If $X(jw)$ is the FT of $x(t)$ then $2\pi x(-jw)$ is the FT of $X(jt)$
  - The IFT of $x(jw)$ is $X(-t)/2\pi$

$$x(jw) = \int_{-\infty}^{\infty} x(t) e^{-jwt} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(jw) e^{jwt} dw$$

$$\int_{-\infty}^{\infty} X(jt) e^{-jw \cdot t} dt$$

$$x(-jw) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(jt) e^{-jwt} dt$$

$$\downarrow$$

$$2\pi x(-jw)$$

Convolution and Multiplication

Time domain	Frequency domain
$x(t)\, y(t)$	$X(jw) * Y(jw) / 2\pi$
$x(t) * y(t)$	$X(jw)Y(jw)$

FT and LTI Systems

- Recall: $h(t) * e^{j\omega t} = e^{j\omega t}\lambda_\omega; \qquad \lambda_\omega = \int h(\tau)e^{-j\omega\tau}d\tau$
- The eigen values of an LTI system are the spectrum of its impulse response
  - $\lambda_w = H(jw)$
- Output of LTI $y(t)$ = IFT{ $Y(jw)$ } = IFT{ $X(jw)\,H(jw)$ } or $Y(jw) = X(jw)\,H(jw)$
  - Frequency response is $\lambda_w = H(jw) = Y(jw) / X(jw)$ for $X(jw) \neq 0$
    - For $x(t) = \delta(t)$, $X(jw) = 1$
  - Bandwidth of $y(t)$ <= minimum{ bandwidth of $x(t)$, bandwidth of $h(t)$}

FT and LTI Systems

- Invertibility:
  - Inverse of $h(t)$ is IFT{ 1 / $H(jw)$ }; non-invertible if $H(jw) = 0$ for some w
- Causality:
  - Any band-limited $H(jw)$ is non-causal (due to uncertainty principle)
  - Vice versa not always true
- Stability:
  - Dirichlet conditions?

Convergence of FT

- Dirichlet conditions:
  - Condition 1: $x(t)$ must be absolutely integrable
  - Condition 2: $x(t)$ must have only a finite number of maxima and minima in any finite interval
  - Condition 3: $x(t)$ must have only a finite number of discontinuities in any finite interval
- These are only sufficient but not necessary conditions
- What about unstable LTI system?

The Ideal LPF

- Find the IFT of the ideal LPF and find its invertibility, causality and stability.

Dirac Delta to the Rescue (Again)

- FT of unit step function is $\pi\delta(w) + 1/jw$ and of a constant is $2\pi\delta(w)$c
  - $\lim_{a \to 0} e^{-at} u(t) = u(t)$ $\int_{-\infty}^{\infty} \frac{a}{a^2 + x^2} dx = \pi$

- Find the FT of sign(t)

FT of Differentiation and Integrator

- $\mathrm{FT}\{\mathrm{d}x(\mathrm{t})/\mathrm{dt}\} = \mathrm{j}wX(\mathrm{j}w)$

- $\mathrm{FT}\{\int \mathrm{x}(\tau)\mathrm{d}\tau\} = \pi X(0)\delta(w) + X(\mathrm{j}w)/\mathrm{j}w$

FT of Periodic Signals

- $\mathrm{FT}\{e^{jst}\} = 2\pi\delta(w - s)$

- Find FT of $\cos(vt)$

Relationship between FS and FT

- For a periodic signal x_p(t),

$$x_p(t) = x(t) * \sum_{n=-\infty}^{\infty} \delta(t - nT) \Rightarrow X_p(j\omega) = X(j\omega)\frac{2\pi}{T}\sum \delta\left(\omega - n\frac{2\pi}{T}\right)$$

$\delta(t - nT)$

$\frac{2\pi}{T}\ \delta(\omega - n\cdot\frac{2\pi}{T})$

- Then, $a_k = X(jk2\pi/T) / T$ = area of $X_p(jk2\pi/T) / 2\pi$ and $X_p(jk2\pi/T) = 2\pi a_k \delta(w - k2\pi/T)$

FT examples

- Find the FT of the triangular signal.

FT examples

- Find the FT of $x(t + T) + x(t - T)$

Solving ODEs

- ODE's are converted to partial fractions by FT

$$\sum_{n=0}^{N} b_n \frac{d^n y(t)}{dt^n} = \sum_{m=0}^{M} c_m \frac{d^m x(t)}{dt^m}$$

$$Y(j\omega) \sum_{n=0}^{N} b_n (j\omega)^n = X(j\omega) \sum_{m=0}^{M} c_m (j\omega)^m$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{m=0}^{M} c_m (j\omega)^m}{\sum_{n=0}^{N} b_n (j\omega)^n}$$

$$= \frac{\sum_{m=0}^{M} c_m (j\omega)^m}{\prod_{n=1}^{N} (d_n + j\omega)} = \sum_{n=1}^{N} \frac{f_n}{d_n + j\omega}$$

- Assuming unique roots

$$h(t) = \sum_{n=1}^{N} f_n e^{-d_n t} u(t)$$

$$a_0 y(t) + a_1 \frac{dy(t)}{dt} + a_2 \frac{d^2 y(t)}{dt^2} + \dots = c_0 x(t) + c_1 \frac{dx(t)}{dt} + \dots$$

$$\sum_{n=0}^{N} a_n \frac{d^n y(t)}{dt^n} = \sum_{m=0}^{M} c_m \frac{d^m x(t)}{dt^m}$$

assume $x(t)$ as input to system and output is $y(t)$

$\Rightarrow$ This is an LTI system

Apply F.T both sides:

$$\sum_{n=0}^{N} a_n (j\omega)^n Y(\omega) = \sum_{m=0}^{M} c_m (j\omega)^m X(\omega)$$

These are polynomials

$$\Rightarrow \frac{Y(\omega)}{X(\omega)} = \frac{\sum_{m=0}^{M} c_m (j\omega)^m}{\sum_{n=0}^{N} a_n (j\omega)^n}$$

$\frac{Y(\omega)}{X(\omega)}$ ↙ frequency responce of the CTI system $H(\omega)$

roots of this polynomial is $d_n = \prod_{n=1}^{N} (j\omega + d_n)$

Now write this as partial fraction

we get $H(\omega) = \sum_{n=1}^{N} \frac{f_n}{d_n + j\omega}$

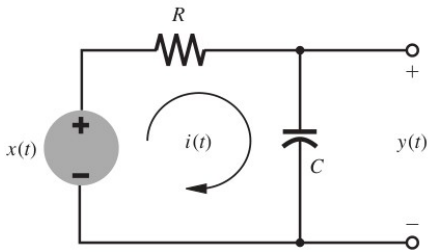
$$\Rightarrow H(t) = \sum_{n=1}^{N} f_n e^{-d_n t} u(t)$$

Solving ODEs

- What if the roots repeat?
  - For a > 0, IFT$\{(a + jw)^{-n}\} = t^{n-1}e^{-at}u(t)/(n-1)!$
- IFT of $dX(jw)/dw$ is $-jt\,x(t)$
- IFT$\{\int X(jv)dv\} = \pi x(0)\delta(t) - x(t)/jt$

Solving ODEs - Example

- **RC circuit**



$$x(t) = \frac{1}{C}\int_{-\infty}^{t} i(v)dv + Ri(t) = y(t) + Ri(t)$$

$$x(t) = \left(1 + RC\frac{d}{dt}\right) y(t)$$

$$X(\omega) = (1 + RC j\omega)\, Y(\omega) \quad \Rightarrow H(\omega) = \frac{1/RC}{1/RC + j\omega}$$

$$\Rightarrow h(t) = 1/RC\, e^{-t/RC}\, u(t)$$

- **RLC circuit** $x(t) = \frac{1}{C}\int_{-\infty}^{t} i(v)dv + L\frac{di(t)}{dt} + Ri(t) = y(t) + LC\frac{d^2y(t)}{dt^2} + RC\frac{dy(t)}{dt}$

FT and RLC circuits

- Impedance in RLC circuits
  - Complex number
    - $Z = a + jwb$
      - $b < 0$: capacitive
      - $b > 0$: inductive
      - $b = 0$: resistive
  - Initial condition, I(0) = 0 → we can set

time-domain

$$v_R(t) = Ri(t)$$

$$v_C(t) = \frac{1}{C}\int_{-\infty}^{t} i(s)ds$$

$$v_L(t) = L\frac{di(t)}{dt}$$

frequency domain

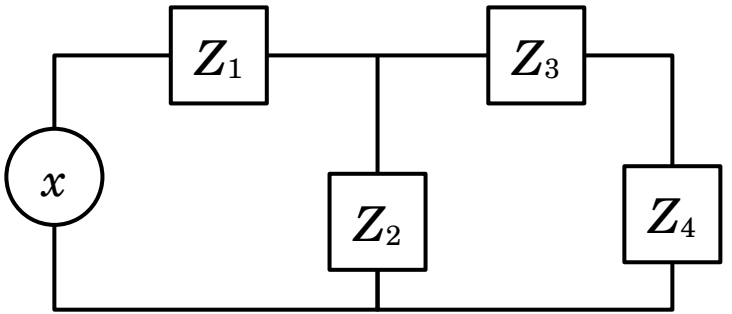
$$V_R(j\omega) = RI(j\omega)$$

$$V_C(j\omega) = \frac{1}{Cj\omega}I(j\omega) + \pi I(0)\delta(\omega)$$

$$V_L(j\omega) = Lj\omega I(j\omega)$$

Electric Circuit Application

- For input voltage $x(t)$, find the output voltage across Z_4 (use KVL), and the frequency response



Sampling Theorem

- Sampling: If $x(t)$ is multiplied with an impulse train, then find the spectrum of this sampled 'discrete' signal

Grand Summary of Fourier Analysis

Summary

- Phasors are eigenfunctions of LTI systems : orthogonal and bounded
- Parseval's theorem: Power/energy in any representation are equal
- Fourier series: Decomposition of a periodic signal in terms of phasors
  - Phasors of a fundamental frequency and its harmonics
  - FS coefficients: Component (projection) of a phasor in the given signal
- Fourier transform: Decomposition of aperiodic (or periodic) in terms of phasors
  - Phasors with a continuum of frequencies

Summary

- Fourier transform of $x(t)$ is $X(jw)$ and the inverse Fourier transform of $X(jw)$ is $x(t)$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

$$E_x = \int |x(t)|^2 dt = \frac{1}{2\pi} \int |X(j\omega)|^2 d\omega = \frac{E_X}{2\pi}$$

- Fourier series of a periodic signal with period T:

$$z(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega t}; \qquad \omega = \frac{2\pi}{T}$$

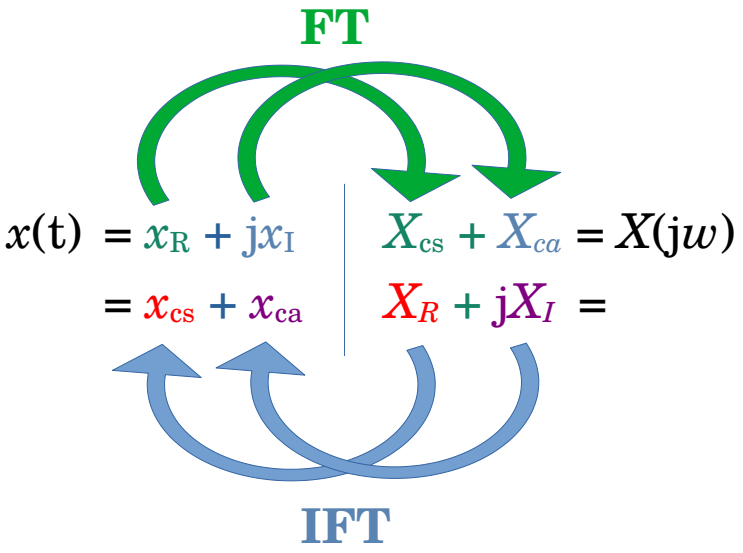
$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} z(t) e^{-jk\omega t} dt$$

  - Energy $E_{z,T} = T \, \Sigma_k |a_k|^2$
  - Power $P_z = E_{z,T}/T = \Sigma_k |a_k|^2$

Summary

- Dirichlet conditions ensure existence of FT and FS (sufficient but not necessary)
- Gibbs phenomenon (both in periodic and aperiodic signals) – occurs at discontinuities
- Uncertainty principle: Finite time (bandwidth) → Infinite bandwidth (time)
- Frequency response is the eigen value of the LTI system corresponding to the phasor of the given frequency
  - It is the ratio of the FT of the output to the FT of input (if the input spectrum ≠ 0)
- Duality: $\mathrm{FT}\{X(\mathrm{j}t)\} = 2\pi x(-\mathrm{j}w)$ and $\mathrm{IFT}\{x(\mathrm{j}w)\} = X(-t)/2\pi$

Summary



Time domain	FT	FS
$x(t)\, y(t)$	$X(jw) * Y(jw) / 2\pi$	$a_k * b_n$
$x(t) * y(t)$	$X(jw)Y(jw)$	

$x(t)$ – Time domain	FT or FS
Even	Even
Odd	Odd
Conj. Sym.	Real
Conj. Asym.	Imaginary
Real	Conj. Sym.
Imaginary	Conj. Asym.

Summary

$x(t)$ – Time domain	a_k – Fourier coefficients	$X(jw)$ – Frequency domain	Bandwidth
$x^*(t)$	a_{-k}^*	$X^*(-jw)$	No change
$x(-t)$	a_{-k}	$X(-jw)$	No change
$x(t-s)$	$a_k e^{-jkws}$ (Only phase changes)	$X(jw)e^{-jws}$	No change (Only phase change in the other domain)
$x(t)e^{jvt}$		$X(j(w-v))$	
$x(ct)$	$a_{k/c}$ (Coefficients are same but fundamental frequency changes)	$X(jw/c) / \lvert c \rvert$ (density changes, area is fixed)	$c > 1$: expands $c < 1$: shrinks

Summary

- Differentiation (HPF)
  - FT{dx(t)/dt} = jwX(jw); IFT{dX(jw)/dw} = -jt x(t)
  - FS{dz(t)/dt} = a_k jk$w$
- Integration (LPF)
  - FT{$\int$x(τ)dτ} = $\pi X(0)\delta(w) + X(jw)/jw$; IFT{$\int X(jv)dv$} = $\pi x(0)\delta(t) - x(t)/jt$
  - FS{$\int^t z(\tau)d\tau$} = a_k/jk$w$
- ODEs can be solved easily by partial fractions and IFT{$(a + jw)^{-n}$} = $t^{n-1}e^{-at}u(t)/(n-1)!$